SEMi-Complete by Design: A Monte Carlo simulation to assess Measurement Invariance in Moderated Nonlinear Factor Analysis and SEM-Trees

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General Information

What is the title of the project?

SEMi-Complete by Design: A Monte Carlo simulation to assess Measurement Invariance in Moderated Nonlinear Factor Analysis and SEM-Trees.

Who are the current and future project contributors?

Leonie Hagitte, Andreas M. Brandmaier

Provide a description of the project.

Ensuring the validity of psychological assessments is crucial, yet differential item functioning (DIF) can threaten measurement invariance (MI) when test items function differently across groups (Bauer, Belzak, and Cole 2020). Recent calls for improved DIF detection methods emphasize the need for more advanced statistical approaches (Lee, Han, and Choi 2024). Moderated nonlinear factor analysis (MNLFA) is a recent approach for assessing MI via parameter moderation within a single-group confirmatory factor analysis framework. MLNFA evaluates MI across multiple continuous and categorical covariates, and accounts for heteroskedasticity by modeling factor and residual variances as functions of these covariates. While MNLFA offers continuous moderation of several parameters of SEMs (e.g.: factor loadings, covariances etc.), it requires a priori specification of covariates and their functional relationships (Bauer 2017; Kolbe et al. 2024). In contrast, structural equation modeling (SEM) trees and forests are data-driven, non-parametric methods that use recursive partitioning to identify latent subgroups in which model parameters differ, without assuming specific functional forms or predefined covariate effects. These approaches allow for nonlinear moderation of factor loadings and can reveal complex interaction effects, enabling the exploratory detection of DIF (Brandmaier et al. 2016,

2013). In this study, we conducted a Monte Carlo simulation to compare the performance of MNLFA and SEM trees and forests in detecting DIF and assessing MI under varying conditions. Specifically, we evaluate their effectiveness in identifying non-invariance and detecting relevant covariates. Our findings will inform best practices for selecting statistical techniques to test MI in psychological assessment.

Did any of the contributors already conduct related simulation studies on this specific question?

Neither of the authors has conducted simulation studies that are of immediate relevance to the current project before. However, Andreas Brandmaier was involved in conducting simulation studies on SEM trees/ SEM forests as well as SEM modeling in general e.g.(Buchberger et al. 2024; Díaz, Heene, and and 2025; Arnold, Voelkle, and Brandmaier 2021).

Aims

What is the aim of the simulation study?

The aim of this simulation study is to evaluate different methods (i.e. MNLFA and SEM trees/ SEM forests) regarding their accuracy, power, and false-positive rate in detecting MI across different factor models (ADEMP category 'hypothesis testing').

Data-Generating Mechanism

How will the parameters for the data-generating mechanism (DGM) be specified?

The data will be generated parametrically. Different population structural equation models (SEM) with latent variables and continuous indicators will be simulated in two different simulation studies:

Study 1

Single-Factor Model with Moderated Parameters

We specify a confirmatory factor analysis model (model 1) with a single latent factor η and four observed indicators $\mathbf{x} = (x_1, \ldots, x_4)^\top$. The influence of continuous moderator variables is introduced into selected parameters via predefined transformation functions applied to underlying raw moderator variables.

The population model is:

$$\mathbf{x} = \boldsymbol{\nu}(Z) + \boldsymbol{\Lambda}_x(Z)\,\eta + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_\varepsilon).$$

where

- $\boldsymbol{\nu}(Z)$ is the vector of intercepts,
- $\boldsymbol{\Lambda}_x(Z)$ is the 4×1 factor loading matrix,
- $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}_\varepsilon)$ is the residual vector,
- $\eta \sim \mathcal{N}\Big(\mu_\eta(Z), \sigma^2_\eta(Z)\Big)$ is the latent factor,
- $Z = h(M)$ is the effective moderator entering the model equations, obtained by transforming a raw continuous covariate M.

Moderator Transformation

Let $M \sim \mathcal{U}(-1, 1)$ denote a bounded continuous covariate.

The effective moderator is defined as

$$Z = h(M),$$

where

$$h(\cdot)$$

is chosen from the following deterministic transformation functions. Each transformation type is treated as a distinct condition in the simulation design:

1. **Linear:**

$$h(M) = M$$

In this case, the effective moderator is bounded to the interval $(-1, 1)$ and symmetric around zero.

2. **Sigmoid:**

$$h(M) = a + \frac{b-a}{1+\exp\Big(-k(M-c)\Big)}, \qquad a = -1,\ b = 1,\ c = 0,\ k > 0.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator symmetric around zero, exhibiting diminishing sensitivity at extreme values.

3. **Quadratic:**

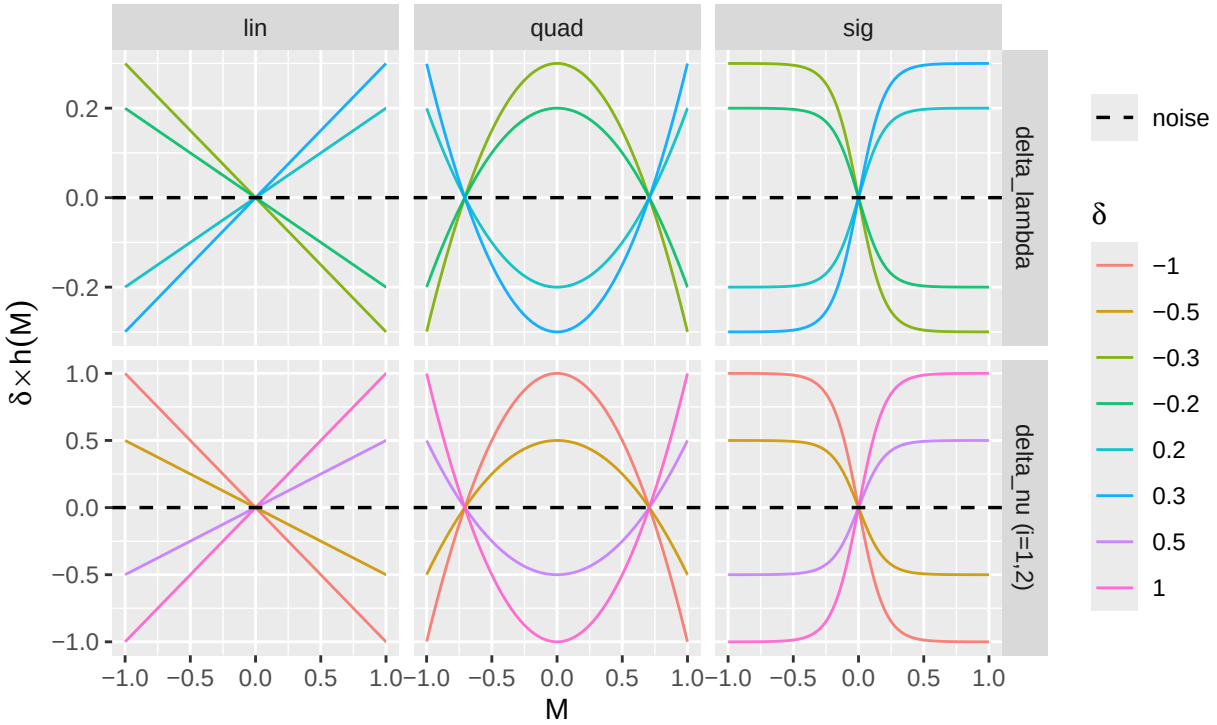
$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. **Noise:**

$$h(M) = 0$$

In this condition, the moderator has no effect on any model parameter and serves as a non-informative (noise) moderator. Null moderation is represented by the noise-moderator condition ($Z = 0$), not by $\delta = 0$; moderation coefficients are held fixed across conditions. All parameter-level moderation effects are specified as linear functions of the effective moderator Z. Nonlinearity in moderation arises exclusively through the transformation $Z = h(M)$, not through nonlinear parameter functions. The transformation $h(M)$ is part of the DGM in every moderation condition.



Note. The dashed horizontal line indicates the null moderation case (noise: delta*h(M) = 0 for all M).
The moderation strength for the latent covariance shares the same delta values as delta_lambda in {−0.3, −0.2, 0.2, 0.3}.

Factor Loadings

The baseline factor loadings were fixed to a discrete value of $\lambda_{xi} = 0.7$.

Variation in loadings across simulation conditions arises exclusively through the presence or absence of moderation, which is treated as a fully crossed factor in the grid design.

Moderated items

At the population level, the baseline (unmoderated) measurement model is specified with a single latent factor and four indicators. The factor loading matrix is constructed such that all four indicators share a common loading value λ:

$$\mathbf{\Lambda}_x = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix}$$

The latent factor variance is fixed at a prespecified constant value $\psi_\eta = 1$ and is held constant across all simulation conditions.

For each condition involving moderation, the loading of item i is defined as a deterministic function of the moderator Z:

$$\lambda_{xi}(Z) = \lambda_{xi} + \delta_\lambda Z, \quad Z \in [-1, 1].$$

The parameter δ_λ represents the strength of the moderation effect and is specified using a discrete set of admissible values chosen to ensure that the moderated loading remains within a psychometrically plausible interval:

$$\lambda_{xi}(Z) \in [0.3,\ 1.0].$$

Given the fixed baseline loading $\lambda_{xi} = 0.7$, this constraint yields an admissible range

$$\delta_\lambda \in [-0.4,\ 0.3].$$

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

For moderator transformations beyond the linear case (quadratic or sigmoid), the functional form of $\lambda_{xi}(Z)$ is adapted accordingly, and the same discrete set of admissible moderation strengths is applied. This yields crossed combinations of (i) moderator

transformation, (ii) moderated parameter type, (iii) number of moderated items, and (iv) moderation strength.

Factor loading matrix

$$\mathbf{\Lambda}_x(Z) = \begin{bmatrix} \lambda_1(Z) \\ \lambda_2(Z) \\ \lambda_3(Z) \\ \lambda_4(Z) \end{bmatrix}$$

Loading moderation is manipulated via the pattern of moderated items and the magnitude of the moderation effect. For each indicator i, the loading is modeled as

$$\lambda_i(Z) = \lambda_i + \delta_\lambda Z,$$

where λ_i denotes the baseline loading and δ_λ controls the strength and direction of loading moderation.

No loading moderation:

$$\delta_\lambda = 0 \quad \text{for all } i = 1, \ldots, 4.$$

Partial loading moderation:

Only the loadings of items 1 and 2 vary with Z:

$$\delta_\lambda \in \{-0.3, -0.2, 0.2, 0.3\} \quad \text{for } i \in \{1, 2\}, \qquad \delta_\lambda = 0 \quad \text{for } i \in \{3, 4\}.$$

Full loading moderation:

All four loadings vary with Z:

$$\delta_\lambda \in \{-0.3, -0.2, 0.2, 0.3\} \quad \text{for } i = 1, \ldots, 4.$$

Residual variances

Item reliabilities are varied deterministically across simulation conditions using the fixed grid

$$Reliability_i \in \{0.60,\ 0.70,\ 0.80,\ 0.95\}.$$

Baseline residual variances are derived analytically from a fixed latent variance $\psi_\eta = \psi_0 > 0$ and indicator-specific reliability values:

$$Reliability_i = \frac{\lambda_i^2 \psi_0}{\lambda_i^2 \psi_0 + \theta_{i0}} \quad \Rightarrow \quad \theta_{i0} = \frac{\lambda_i^2 \psi_0 (1 - \text{Reliability}_i)}{\text{Reliability}_i}.$$

The residual variances θ_{i0} are fixed at these baseline values across moderator values and population models. Consequently, when factor loadings are moderated, reliability may vary as a function of the moderator.

Intercepts

Intercept moderation is manipulated via the pattern of moderated items and the magnitude of the moderation effect. For each indicator i, the intercept is modeled as

$$\nu_i(Z) = \nu_i + \delta_\nu Z,$$

where ν_i denotes the baseline intercept and δ_ν controls the strength and direction of intercept moderation.

No intercept moderation:

$$\delta_\nu = 0 \quad \text{for all } i = 1, \ldots, 4.$$

Partial intercept moderation:

Only the intercepts of items 1 and 2 vary with Z:

$$\delta_\nu \in \{-1.0, -0.5, 0.5, 1.0\} \quad \text{for } i \in \{1, 2\}, \qquad \delta_\nu = 0 \quad \text{for } i \in \{3, 4\}.$$

Full intercept moderation:

All four intercepts vary with Z:

$$\delta_\nu \in \{-1.0, -0.5, 0.5, 1.0\} \quad \text{for } i = 1, \ldots, 4.$$

Baseline intercepts are set to 1 across simulation conditions.

Population model overview

The population models are generated by combining the loading- and intercept-moderation specifications above.

Model	Loading moderation	Intercept moderation	Active moderator structure
0	none	none	none
1.1	full	none	$Z_1 = h(M_1)$
1.11	none	full	$Z_1 = h(M_1)$
1.12	full	full	$Z_1 = h(M_1)$
1.2	partial, items 1–2	none	$Z_1 = h(M_1)$
1.21	none	partial, items 1–2	$Z_1 = h(M_1)$
1.22	partial, items 1–2	partial, items 1–2	$Z_1 = h(M_1)$
1.3	full	none	$Z_1 = h(M_1)$, $Z_2 = h(M_2)$, and $Z_1 Z_2$

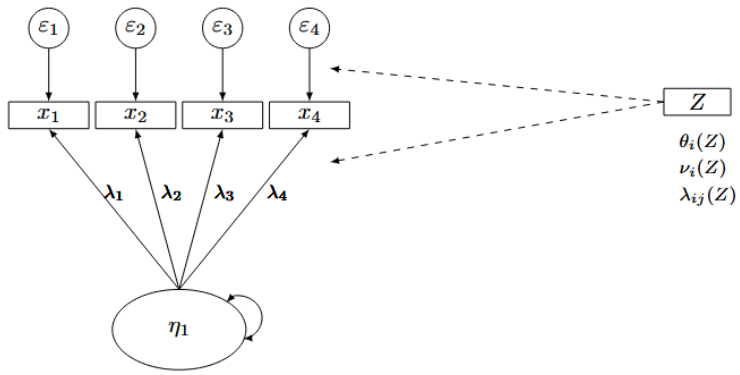
For Model 1.3, the loading equation is

$$\lambda_i(Z_1, Z_2) = \lambda_i + \delta_{\lambda 1} Z_1 + \delta_{\lambda 2} Z_2 + \delta_{\lambda 12} Z_1 Z_2, \quad i = 1, \ldots, 4.$$

Latent distribution

The latent variance is fixed at 1.

Analytical Model Study 1



The analytical model will contain one latent factor η_1, with four manifest indicators $x_1, ..., x_4$, their factor loadings $\lambda_1, ..., \lambda_4$ and their respective residuals $\epsilon_1, ..., \epsilon_4$. In the analytical model the residuals will be assumed and modeled to be uncorrelated. For MNLFA, moderation of selected intercepts and factor loadings is modeled parametrically. In the main study, two MNLFA specifications will be evaluated: one assuming linear moderation in the observed moderator(s), and one assuming quadratic moderation. Therefore, MNLFA is correctly specified only when the analytical functional form matches the DGM transformation $h(M)$, and is otherwise intentionally misspecified. For most population models, the MNLFA includes M_1 as the focal moderator. In the interaction population model, an additional MNLFA specification includes M_1, M_2, and $M_1 \times M_2$.

Study 2

Two Factors with Moderator Effects

We specify a confirmatory factor analysis model with two latent factors, η_1 and η_2, and seven observed indicators $\mathbf{x} = (x_1, \ldots, x_4)^\top$ and $\mathbf{y} = (y_1, \ldots, y_3)^\top$. Along their respective residuals $\epsilon_1, ..., \epsilon_4$ and $\delta_1, ..., \delta_3$. The first factor loads onto indicators x_1 to x_4, the second onto y_1 to y_3. A continuous moderator variable Z introduces conditional heterogeneity into selected measurement parameters (i.e. intercepts, loadings and latent

factor covariance) using one of three functional forms (linear, quadratic or sigmoid).

$$\mathbf{x} = \boldsymbol{\nu}_x(Z) + \boldsymbol{\Lambda}_x(Z)\,\eta_1 + \boldsymbol{\varepsilon}(Z),$$

$$\mathbf{y} = \boldsymbol{\nu}_y(Z) + \boldsymbol{\Lambda}_y(Z)\,\eta_2 + \boldsymbol{\delta}(Z),$$

where:

- $\boldsymbol{\nu}_x(Z)$ and $\boldsymbol{\nu}_y(Z)$ are the 4×1 and 3×1 vectors of intercepts for $\mathbf{x}$ and $\mathbf{y}$, respectively,
- $\boldsymbol{\Lambda}_x(Z)$ is the 4×1 loading vector for η_1, and $\boldsymbol{\Lambda}_y(Z)$ is the 3×1 loading vector for η_2,
- $\boldsymbol{\varepsilon}(Z) \sim \mathcal{N}\Big(\mathbf{0}, \boldsymbol{\Theta}_x(Z)\Big)$ and $\boldsymbol{\delta}(Z) \sim \mathcal{N}\Big(\mathbf{0}, \boldsymbol{\Theta}_y(Z)\Big)$ are the residual vectors for $\mathbf{x}$ and $\mathbf{y}$,
- $\boldsymbol{\eta} = (\eta_1, \eta_2)^\top \sim \mathcal{N}\Big(\boldsymbol{\mu}_\eta(Z), \boldsymbol{\Phi}_\eta(Z)\Big)$ is the latent factor vector,
- $Z = h(M)$ is the effective moderator entering the model equations, obtained by transforming a raw continuous covariate M.

Moderator Transformation

Let $M \sim \mathcal{U}(-1, 1)$ denote a bounded continuous covariate.

The effective moderator is defined as

$$Z = h(M),$$

where

$$h(\cdot)$$

is chosen from the following deterministic transformation functions. Each transformation type is treated as a distinct condition in the simulation design:

1. **Linear:**

$$h(M) = M$$

In this case, the effective moderator is bounded to the interval $(-1, 1)$ and symmetric around zero.

2. **Sigmoid:**

$$h(M) = a + \frac{b-a}{1+\exp\Big(-k(M-c)\Big)}, \qquad a = -1,\ b = 1,\ c = 0,\ k > 0.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator symmetric around zero, exhibiting diminishing sensitivity at extreme values.

3. **Quadratic:**

$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. **Noise:**

$$h(M) = 0$$

In this condition, the moderator has no effect on any model parameter and serves as a non-informative (noise) moderator.

Null moderation is represented by the noise-moderator condition ($Z = 0$), not by $\delta = 0$; moderation coefficients are held fixed across conditions. All parameter-level moderation effects are specified as linear functions of the effective moderator Z. Nonlinearity in moderation arises exclusively through the transformation $Z = h(M)$, not through nonlinear parameter functions.

Factor Loadings

For each latent factor, all baseline loadings for that factor are fixed at 0.7. For the two-factor model in Study~2, this yields

$$\mathbf{\Lambda}_x = \begin{bmatrix} 0.7 \\ 0.7 \\ 0.7 \\ 0.7 \end{bmatrix}, \qquad \mathbf{\Lambda}_y = \begin{bmatrix} 0.7 \\ 0.7 \\ 0.7 \end{bmatrix},$$

Moderation of factor loadings is introduced only for the first two x-indicators. For each moderated indicator $i \in \{1, 2\}$,

$$\lambda_{xi}(Z) = \lambda_{xi} + \delta_\lambda Z.$$

To ensure that moderated loadings remain within an admissible interval,

$$\lambda_{xi}(Z) \in [0.3,\ 1.0],$$

the moderation coefficients are restricted to

$$\delta_\lambda \in [-0.4,\ 0.3],$$

and evaluated on the discrete grid (leaving out the zero effect)

$$\delta_\lambda \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

Thus, the loading matrices take the form

$$\mathbf{\Lambda}_x(Z) = \begin{bmatrix} \lambda_{x1}(Z) \\ \lambda_{x2}(Z) \\ 0.7 \\ 0.7 \end{bmatrix}, \qquad \mathbf{\Lambda}_y(Z) = \begin{bmatrix} 0.7 \\ 0.7 \\ 0.7 \end{bmatrix},$$

yielding fully crossed combinations of (i) moderator transformation type $Z = h(M)$, (ii) moderated vs. unmoderated indicators, and (iii) moderation strength δ_λ.

Residual Variances

Item reliabilities are varied deterministically across simulation conditions using the fixed grid.

$$Reliability_i \in \{0.60,\ 0.70,\ 0.80,\ 0.95\}$$

Baseline residual variances are derived analytically from a fixed latent variance $\psi_\eta = \psi_0 > 0$ and indicator-specific reliability values:

$$Reliability_i = \frac{\lambda_i^2 \psi_0}{\lambda_i^2 \psi_0 + \theta_{i0}} \quad \Rightarrow \quad \theta_{i0} = \frac{\lambda_i^2 \psi_0 (1 - \text{Reliability}_i)}{\text{Reliability}_i}.$$

The residual variances θ_{i0} are fixed at these baseline values across moderator values and population models. Consequently, when factor loadings are moderated, reliability may vary as a function of the moderator.

Intercepts

Intercept moderation is manipulated via two fully crossed factors:

(i) the pattern of moderated items and

(ii) the magnitude of the moderation effect. For each indicator i, the intercept is modeled as

$$\nu_i(Z) = \nu_i + \delta_\nu Z,$$

where ν_i denotes the baseline intercept and δ_ν controls the strength and direction of intercept moderation. The pattern of moderated items is varied across three levels:

No intercept moderation: $\delta_\nu = 0$ for all $i = 1, \ldots, 4$, such that

$$\nu_i(Z) = \nu_i \quad \text{for all } i.$$

Partial intercept moderation (items 1 and 2):

$$\delta_\nu \in \{-1.0,\ -0.5,\ 0.5,\ 1.0\} \quad \text{for } i \in \{1, 2\}, \qquad \delta_\nu = 0 \quad \text{for } i \in \{3, 4\},$$

such that only $\nu_1(Z)$ and $\nu_2(Z)$ vary with Z.

Full intercept moderation (all items):

$$\delta_\nu \in \{-1.0,\ -0.5,\ 0.5,\ 1.0\} \quad \text{for } i = 1, \ldots, 4,$$

such that the intercepts of all four indicators vary with Z. Baseline intercepts are set to 1.

Latent Variable Distribution

Latent means are held constant at zero for both latent factors:

$$\boldsymbol{\mu}_\eta(Z) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Latent variances are allowed to vary with the effective moderator Z through the log-linear specification

$$\sigma^2_{\eta_j}(Z) = \exp(\beta_{0j} + \beta_{1j} Z)\,, \qquad j = 1, 2,$$

ensuring positivity for all variance values.

The baseline variance is set to 1 and moderation strength is selected from a discrete grid:

$$\delta_{\sigma_\eta} \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

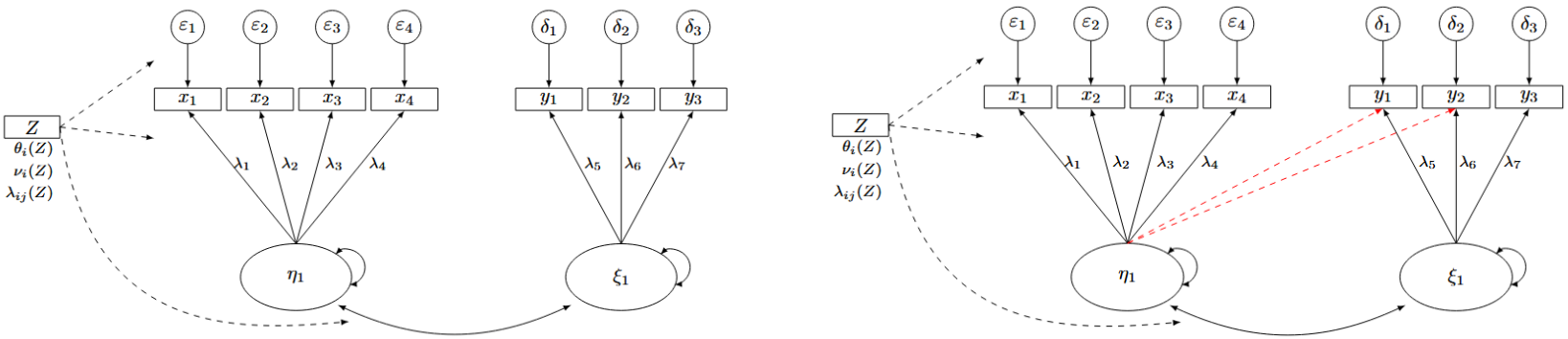
The latent covariance between η_1 and η_2 is fixed across all conditions:

$$\phi_{12}(Z) = 0.4,$$

yielding the latent covariance matrix

$$\mathbf{\Phi}_{\eta}(Z) = \begin{bmatrix} \sigma^2_{\eta_1}(Z) & 0.4 \\ 0.4 & \sigma^2_{\eta_2}(Z) \end{bmatrix}.$$

Analytical Models Study 2



(a) *Model 2.0* (b) *Model 2.1*

Figure 1

We specify a confirmatory factor analysis model with two latent factors, η_1 and ξ_1, and seven observed indicators, $\mathbf{x} = (x_1, \ldots, x_4)^\top$ and $\mathbf{y} = (y_1, y_2, y_3)^\top$, along with their respective residuals, $\varepsilon_1, \ldots, \varepsilon_4$ and $\delta_1, \delta_2, \delta_3$. The first factor, η_1, loads onto indicators x_1 to x_4, while the second factor, ξ_1, loads onto indicators y_1 to y_3. A continuous moderator variable Z introduces conditional heterogeneity into selected measurement parameters, including intercepts, factor loadings, and latent factor covariance, using one of two functional forms (linear or quadratic). Red dashed arrows indicate model misspecifications.

What will be the different factors of the data-generating mechanism?

The simulation design includes several manipulated factors that govern the data-generating mechanism (DGM). These factors are varied across simulation conditions and apply consistently to both studies, unless otherwise noted.

Core DGM Factors (Applied in Both Studies)

The following factors are systematically varied in both studies using a grid search design:

Functional form of moderation. The effective moderator $Z = h(M)$ is evaluated systematically using the following DGM functions:

1. Linear
2. Quadratic
3. Sigmoid
4. Noise/null moderation, $h(M) = 0$

Strength of moderation. In each case where parameters are subjected to moderation, the strength of the moderation δ is varied:

δ_λ is evaluated at the following discrete levels:

$$\delta_\lambda \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

$$\delta_\nu \in \{-1.0,\ -0.5,\ 0.5,\ 1.0\} \quad \text{for } i \in \{1, 2\}, \qquad \delta_\nu = 0 \quad \text{for } i \in \{3, 4\},$$

Model misspecification of moderator form. The DGM always defines the effective moderator through $Z = h(M)$. In MNLFA, the analytical model is fitted using

prespecified parametric forms, namely linear and quadratic moderation. Consequently, MNLFA is correctly specified when the fitted analytical form corresponds to the DGM transformation and misspecified otherwise. SEM trees do not impose a parametric functional form, but they must identify the correct moderator variable(s) from the candidate set.

Study-Specific Factors

Study 1. Focuses on functional form misspecification and data-related conditions:

Sample sizes: $N \in \{300, 500, 700, 1000\}$ **Intercepts:** Intercept baselines are set to 1 across simulation conditions.

Indicator reliabilities: *Low* (.6), *moderate* (.7), *moderate to high* (.8), or *high* (.95).

Study 2. Builds on Study 1 and introduces additional model misspecification in the factor structure:

Structural model misspecifications: (between-subjects factor) **1. No misspecification:** Factor structure matches the analysis model. **2. Cross-loadings:** Items y_5, y_6 additionally load on a second latent factor:

Latent Covariance: The latent covariance is moderated by Z. The baseline variance is set to 1 and moderation strength is selected from:

$$\delta_{\sigma_\eta} \in \{-0.3,\ -0.2,\ 0.2,\ 0.3\}.$$

If there is more than one factor: How will the factor levels be combined and how many simulation conditions will this create?

We employ a full grid search, in which parameter values are sampled from pre-specified discrete values. A total of 6000 simulation replications should ideally be conducted. This number was chosen to ensure that the confidence intervals for key

estimated proportions (e.g., power or Type I error rates near $p = 0.8$ achieve an approximate width of $\pm 1\%$. This is based on the calculation:

$$\text{SE} = 2 \times \sqrt{\frac{p(1-p)}{6000}},$$

which yields a standard error consistent with the desired precision for evaluating simulation outcomes within this study. To ensure computational feasibility, we propose an iterative approach whereby initial pilot simulations $n = 100$ are conducted to amongst others, estimate the needed simulation time. These estimates then inform a more principled determination of the total number of replications, targeting a predefined precision threshold.

The factors that vary across simulation conditions include:

- The functional form of moderator effects (linear, quadratic, sigmoid)
- The type of moderated parameters (e.g., factor loadings, intercepts)
- The degree of model misspecification (none, cross-loadings, correlated residuals, combined)
- Item reliabilities (ranging from 0.6 to 0.95)
- Moderator strength parameters
- Sample sizes per group (i.e. 300, 500, 700, 1000)

Estimands and Targets

What will be the estimands and/or targets of the simulation study?

The primary estimands in this simulation study pertain to the detection and accurate estimation of measurement non-invariance (MI) due to moderation effects on measurement parameters. Specifically, we assess whether the estimation methods compared can correctly identify the presence or absence of moderation in factor loadings, intercepts,

and residual variances, reflecting the true data-generating structure. Consequently, key simulation outcomes include Type I error rates and statistical power associated with detecting such moderation effects, as well as bias and variability in the recovered parameter estimates.

Secondary estimands include model-level indicators of fit (e.g., Kullback–Leibler divergence, AIC), which serve as auxiliary diagnostics for methods that yield such metrics (i.e., MNLFA). These are not to the same extent applicable to tree-based approaches and are thus interpreted as method-specific targets rather than general evaluative criteria.

Methods

How many and which methods will be included and which quantities will be extracted?

We will compare the following methods:

1) **Moderated Nonlinear Factor Analysis:** MNLFA was initially introduced by Bauer and Hussong (2009) within the framework of integrative data analysis and was subsequently extended into a general approach for assessing measurement invariance and moderation (Bauer 2017). The fundamental principle of MNLFA is the parameterization of both measurement and structural model parameters as functions of one or more moderator variables. MNLFA accommodates both frequentist estimation methods, such as maximum likelihood (ML), and Bayesian techniques, including Markov Chain Monte Carlo (MCMC), thereby offering methodological flexibility aligned with specific analytical objectives (Münch and Koch 2024).

2) **SEM Trees:** Structural equation modeling (SEM) trees were originally introduced by Brandmaier et al. (2013) as a way to systematically search for important covariates and their interactions in data, creating homogeneous subgroups. The method was refined later on by Brandmaier et al. (2016) and Arnold, Voelkle, and

Brandmaier (2021). SEM trees and forests are data-driven, non-parametric methods that build upon the decision tree paradigm, thus using recursive partitioning to identify latent subgroups in which model parameters differ, without assuming specific functional forms or predefined covariate effects. These approaches allow for nonlinear moderation of factor loadings and can reveal complex interaction effects, enabling the exploratory detection of DIF.

For the parametric MNLFA, we will extract estimated moderated parameters (e.g., factor loadings, intercepts), their associated standard errors, and two-sided p-values for testing the null hypothesis of no moderation effect. Additionally, model fit indices such as Kullback–Leibler divergence, AIC and the LRT-statistics will be recorded.

For the parameter estimates and fit indices, we will summarize their distributions across simulation replications by reporting key quantiles (2.5th, 50th, and 97.5th percentiles). The null hypothesis will be rejected if the associated p-value is less than the conventional significance level of 0.05.

In contrast, non-parametric Tree-based approaches do not yield direct parameter estimates with standard errors. We use score-based SEM-trees(Arnold, Voelkle, and Brandmaier 2021). Their detection of MI/ non-invariance is operationalized via the presence of at least one split in the correct condition at the correct respective level of focus parameters. Those splits, introducing a subgroup, will be evaluated at the significance level: 0.05, using the “maximally selected likelihood-ratio”(Arnold, Voelkle, and Brandmaier 2021). Additionally, a Bonferroni correction to account for multiple comparisons at each node. The trees will be allowed to split on the focus parameters relevant for the invariance testing (the means of the intercepts, factor loadings and the latent covariance in study 2). Candidate subgroup structures are assessed by testing whether they produce statistically significant differences in the specified model parameters across groups. Differences in factor variances between subgroups are permitted in the

model, but they are not used as criteria for selecting or evaluating covariates. Furthermore, pre-pruning will be applied, limiting the maximum depth of the tree to 3 and `min.N` is set to 50. Because the MNLFA also considers directly the form of the moderation, which the SEM trees do not, we further want to use SEM forests for generating partial dependence plots to get an approximation of the functional form of the moderating variables.

With this distinction we aim to ensure that each method is evaluated according to its own inferential framework, while maintaining comparability in terms of its ability to detect MI reliably.

Performance Measures

Which performance measures will be used?

For the MNLFA **exclusively**, we will evaluate the following performance measures:

Bias: The average difference between the estimated parameter $\hat{\theta}$ and the true parameter θ, defined as

$$\widehat{\text{Bias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i - \theta,$$

where n_{sim} is the number of simulation replications.

Absolute Bias:

$$\widehat{\text{AbsBias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \left|\hat{\theta}_i - \theta\right|.$$

Relative Bias: (expressed as a proportion of the true parameter)

$$\widehat{\text{RelBias}} = \frac{\widehat{\text{Bias}}}{|\theta|} \quad \text{for } \theta \neq 0.$$

Root Mean Squared Error (RMSE):

$$\widehat{\text{RMSE}} = \sqrt{\frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta)^2}.$$

Coverage Probability: The proportion of confidence intervals that contain the true parameter θ.

Model Fit Indices: (e.g., RMSEA) will be reported if applicable, computed according to their standard definitions.

For the SEM trees **as well as** for the MNLFA we will evaluate:

Type I Error Rate and Power: The rejection rate of the null hypothesis at significance level $\alpha = 0.05$, calculated as

$$\widehat{\text{RRate}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbf{1}(p_i \leq 0.05),$$

where $\mathbf{1}(\cdot)$ is the indicator function.

Performance measures calculated for multiple parameters (e.g., factor loadings) will be aggregated by reporting the mean value across parameters.

How will Monte Carlo uncertainty of the estimated performance measures be calculated and reported?

We will quantify Monte Carlo uncertainty using Monte Carlo Standard Errors (MCSEs) for each performance measure, calculated as follows:

The **Rejection Rate** refers to the proportion of simulated datasets in which a null hypothesis is rejected, commonly used to estimate empirical Type I error rates or statistical power. The **MCSE for the Rejection Rate** is calculated assuming a binomial distribution of rejections across n_{sim} simulation replications, which reflects the standard error of a proportion estimator:

$$\text{MCSE}_{\widehat{\text{RRate}}} = \sqrt{\frac{\widehat{\text{RRate}}(1 - \widehat{\text{RRate}})}{n_{\text{sim}}}},$$

For **Bias**, MCSE is calculated based on the sample variance of the specific parameter estimates across simulation replications:

$$\text{MCSE}_{\widehat{\text{Bias}}} = \frac{S_{\hat{\theta}}}{\sqrt{n_{\text{sim}}}},$$

where

$$S_{\hat{\theta}} = \sqrt{\sum_{i=1}^{n_{\text{sim}}} \{\hat{\theta}_i - (\sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_i / n_{\text{sim}})\}^2 / (n_{\text{sim}} - 1)}$$

is the sample standard deviation of the effect estimates.

Siepe et al. (2024) ,Morris, White, and Crowther (2019) and Whitehead (1993)

How many simulation repetitions will be used for each condition?

A total of 6000 simulation replications should ideally be conducted. This number was chosen to ensure that the confidence intervals for key estimated proportions (e.g., power or Type I error rates near $p = 0.8$) achieve an approximate width of $\pm 1\%$. This is based on the calculation:

$$\text{SE} = 2 \times \sqrt{\frac{p(1-p)}{6000}},$$

which yields a standard error consistent with the desired precision for evaluating simulation outcomes within this study. To ensure computational feasability, we propose an iterative approach whereby initial pilot simulations $n = 100$ are conducted to amongst others, estimate the needed simulation time. These estimates then inform a more principled determination of the total number of replications, targeting a predefined precision threshold.

How will missing values due to non-convergence or other reasons be handled?

Non-convergence or other missing values may occur during model estimation. We plan to:

- Exclude only the specific simulation replicates that fail to converge for a given method while retaining data from other methods within the same replication.
- Report the proportion of non-converged cases per method and condition.
- If the proportion of non-convergence exceeds a pre-specified threshold (e.g., 5%), we will flag the corresponding simulation condition and interpret results with caution.

No imputation of missing estimates will be performed.

How do you plan on interpreting the performance measures?

Answer:

Other

Which statistical software/packages do you plan to use?

Answer:

Which computational environment do you plan to use?

The simulation study will be conducted in R (R Core Team 2025).

Which other steps will you undertake to make simulation results reproducible?

We are planning to make the code of this project publicly available. Software and packages will be documented via "renv"(Ushey and Wickham 2026) and Docker. Furthermore, we are planning to use the "repro" package, as well as the "reproducibleRchunks" package for better reproducibility.

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